

Perception Challenge: Rim Center Identification

This take-home challenge is designed to assess your problem-solving approach and your knowledge of computer vision principles and techniques. Your task is to create a detailed design plan and implementation for a system that finds the center of car wheels in arbitrary orientations.

As a deliverable we would like a document detailing your approach and some code showing off your approach. In order to save you time, a full and complete pipeline implementation is not required however we would like to see some code that demonstrates the techniques you would apply to the exercise. Use your own discretion on selecting what portion you choose to implement - our goal is for you to have fun and show off your skills to us.

We hope you will spend enough time on this project to demonstrate your skills, and we consider 2-4 hours nominal, but use your own judgement. Please try to get back to us within a week of receiving the challenge.

If you have any questions about the challenge or the process, please feel free to reach out.

What we're looking for

We are looking to understand:

- How you approach problems
- The breadth/depth of your knowledge, and your ability to apply it to a specific problem
- How you select a methodology and approach
- Thought process around iteration and trial and error
- Your ability to communicate with peers

You are welcome to submit any form of writing, and we aren't evaluating on how it's presented - feel free to do whatever you prefer! If it is helpful, you can respond directly in this document, share a GitHub repo or whatever else you'd prefer. Please be sure to email us when you are ready for us to take a look.

Please make sure to include the following information:

- What approach you would use to solve the problem, including details on specific methods and tools
 - Include why you have chosen to approach the problem the way you did
- Any additional information or data collection that you would request before actually implementing your approach
- How you would approach testing, validation and reliability of your solution
- Edge cases or subtle details that may come up and your approach for handling them
- Sensor selection, environmental constraints necessary, etc to achieve good results.
- Realistic estimates of implementation time/complexity/risk for your approach

Feel free to assume the audience is well-versed in computer vision.

Problem Description

You are tasked with identifying wheel centers from images. Assume that you have a camera that captures close-up images of a car at 15 frames per second (see examples below). If you wish, you are free to propose an alternative solution with a different sensor modality, using the examples below as a reference. In either case please provide a detailed system design proposal.

You should design a system that does the following:

- If there is a wheel visible in frame, return the pixel position of the center of the wheel
- If there is no wheel in the image, indicate as such

To aid you in this task, we have provided an example set of 1000 labeled images. 64 of these are shown below, with the wheel centers shown in red.

Note that in the example dataset, the wheel center is defined as the center of the ellipse defined by projecting the tire's inner edge into the camera plane. Or in other words, the wheel center point is the intersection of the axis of rotation of the wheel and the outermost plane of the wheel.

Thus in images where the wheel is tilted to the camera, it is slightly offset from the marking on the center of the car's wheel cover. If you use a different definition of center, please note what it is.

